

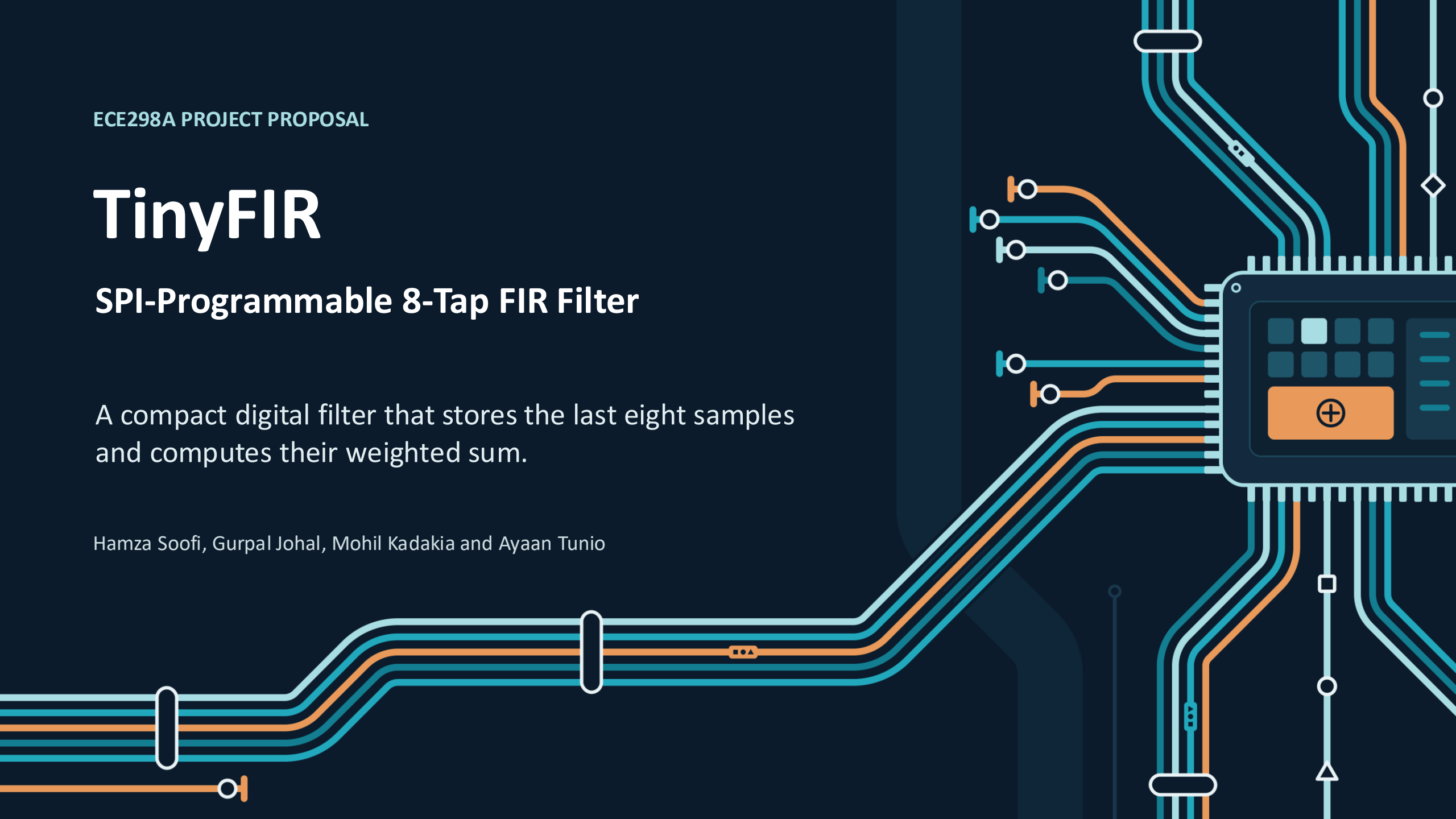
ECE298A PROJECT PROPOSAL

TinyFIR

SPI-Programmable 8-Tap FIR Filter

A compact digital filter that stores the last eight samples and computes their weighted sum.

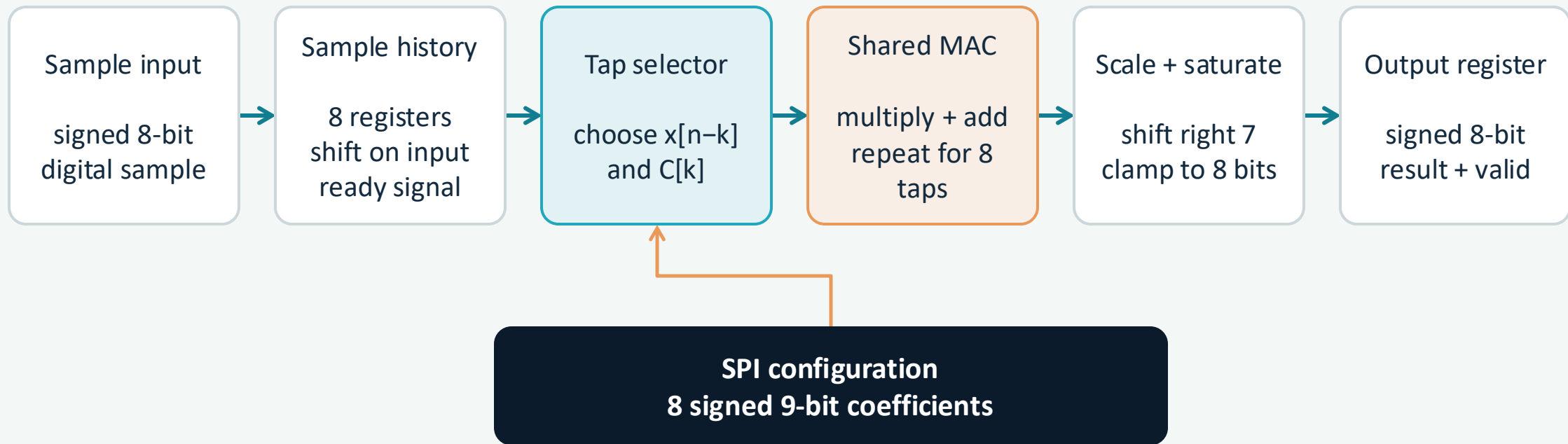
Hamza Soofi, Gurpal Johal, Mohil Kadakia and Ayaan Tunio



Filter architecture

One accepted sample triggers eight sequential tap calculations

$$y[n] = \sum_{k=0}^7 h[k]x[n-k]$$



Targets, verification & team

10 MHz

proposed system clock

≤ 16 cycles

processing latency target

500 ksample/s

maximum sample-rate target

Verification plan

Python integer reference model

Compare RTL against 10,000 random samples. Check reset, SPI writes, impulse response, unity pass-through, signed rounding, and saturation . Demonstrate low-pass, high-pass and band-pass responses.

Work split

Ayaan Tunio

FIR arithmetic datapath

Gurpal Johal

SPI and coefficient storage

Mohil Kadakia

Sample path, control and integration

Hamza Soofi

Python model and verification

Signed 8-bit samples / Signed 9-bit coefficients / 20-bit Accumulator

GF180, 1 × 2 Tiny Tapeout block. Area and timing must pass the course implementation checks.

Eight taps support broad filter demonstrations; sharp frequency transitions are not promised.